

HAMAMATSU, Japan

WMO: 476540

Lat: 34.7539N

Lon: 137.7117E

Elev: 156

StdP: 14.61

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	32.1	33.8	11.7	10.0	40.1	14.5	11.5	40.5	22.4	44.1	20.8	44.0	9.5	290	0.403

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	11.6	91.6	78.1	89.1	77.8	87.2	77.3	80.5	87.0	79.6	85.6	78.8	84.4	11.1	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
78.8	150.4	83.7	78.0	146.4	83.1	77.2	142.5	82.4	44.2	87.4	43.3	85.4	42.5	84.6	83.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)
(4)	19.9	18.0	16.2	DB	28.7	96.8	2.1	2.5	27.3	98.6	26.1	100.0	24.9	101.4	23.4	103.2
(5)				WB	24.4	81.9	1.9	1.2	23.1	82.7	22.0	83.5	20.9	84.2	19.6	85.1

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	62.5	43.4	44.8	50.4	59.0	66.8	72.6	79.9	82.3	76.8	67.7	57.4	47.9
	DBStd	14.04	4.16	4.90	5.19	5.05	3.65	3.72	4.05	3.18	4.37	4.61	5.13	5.12
	HDD50	538	210	159	60	3	0	0	0	0	0	0	4	102
	HDD65	2684	671	566	453	186	22	1	0	0	0	24	231	530
	CDD50	5107	4	13	72	274	522	679	926	1002	805	547	226	37
	CDD65	1780	0	0	0	7	79	230	461	537	355	107	4	0
	CDH74	14072	0	0	0	15	225	947	4165	5743	2632	342	3	0
	CDH80	4218	0	0	0	0	16	152	1303	2057	653	37	0	0
Wind	WSAvg	7.6	9.5	9.3	9.0	8.2	7.2	6.4	6.3	6.4	6.4	6.5	7.2	8.8
Precipitation	PrecAvg	73.4	2.6	3.5	5.7	6.9	7.7	8.7	7.8	5.5	10.1	7.4	4.9	2.6
	PrecMax	100.1	6.3	8.9	9.7	13.5	12.0	14.4	16.7	16.3	25.4	27.1	9.3	5.6
	PrecMin	51.7	0.0	0.9	2.4	1.1	3.3	2.0	0.9	0.9	3.0	1.1	0.6	0.2
	PrecStd	13.1	1.6	2.0	1.7	3.0	2.6	2.7	4.2	3.3	5.2	4.7	2.4	1.6
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	58.8	63.7	69.1	76.1	82.2	87.8	95.1	96.4	91.0	83.8	73.7	65.6
		MCWB	50.8	55.5	55.9	61.6	66.6	74.1	77.8	78.7	76.9	71.4	63.3	58.2
	2%	DB	55.5	59.5	65.2	72.7	78.9	84.0	91.0	92.4	88.0	80.3	70.8	61.8
		MCWB	47.1	51.1	54.2	60.1	65.0	72.4	78.0	78.4	75.9	68.9	60.9	54.0
	5%	DB	53.1	56.2	62.4	70.1	76.4	81.2	88.4	89.9	85.9	77.6	68.4	59.0
		MCWB	44.8	47.5	53.7	59.2	64.2	71.4	77.8	78.4	75.7	67.9	59.8	50.8
	10%	DB	50.9	53.6	59.9	67.9	74.2	79.0	86.3	88.1	83.8	75.4	66.0	56.4
		MCWB	43.1	45.5	51.6	58.3	63.5	70.5	77.3	78.1	74.7	66.5	58.0	48.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	54.8	58.9	61.0	67.8	71.8	77.9	81.5	81.9	80.2	75.2	67.8	61.2
		MCDB	56.2	61.3	62.5	69.5	75.3	84.1	88.7	89.6	85.6	77.9	69.7	63.3
	2%	WB	49.6	53.1	58.4	64.8	69.6	76.0	80.1	80.8	78.9	72.5	64.0	55.7
		MCDB	53.4	57.8	62.0	68.6	73.4	80.3	86.5	87.7	83.6	77.4	67.4	60.0
	5%	WB	46.4	49.9	56.0	62.7	68.0	74.5	79.2	80.0	77.9	70.3	61.7	52.5
		MCDB	50.9	54.1	60.5	67.1	72.6	78.4	84.9	86.4	82.7	75.6	66.0	56.8
	10%	WB	44.2	47.0	53.4	60.7	66.4	72.9	78.4	79.2	76.8	68.4	59.7	50.1
		MCDB	49.8	51.8	58.6	65.9	71.5	76.8	83.7	85.2	81.8	73.6	64.5	55.4
Mean Daily Temperature Range	5% DB	MDBR	12.8	14.1	15.0	14.8	13.6	11.2	10.7	11.6	11.9	12.8	13.3	12.8
		MCDBR	15.1	16.6	17.5	17.0	16.7	14.1	13.7	13.7	13.5	14.4	15.1	15.1
		MCWBR	10.5	11.9	12.3	10.4	9.0	6.4	5.3	5.3	6.4	7.9	9.5	10.5
	5% WB	MCDBR	13.4	14.6	15.0	13.2	12.9	10.6	11.7	11.9	11.5	12.4	12.7	13.2
		MCWBR	11.5	12.8	13.2	10.3	8.9	5.7	5.1	5.2	6.2	8.3	10.4	11.4
Clear-Sky Solar Irradiance	taub	0.375	0.420	0.479	0.525	0.535	0.557	0.548	0.528	0.486	0.460	0.409	0.375	
	taud	2.261	2.139	1.987	1.919	1.967	1.982	2.058	2.104	2.193	2.183	2.243	2.277	
	Ebn at Noon	257	257	251	246	245	239	240	242	246	241	242	248	
	Edh at Noon	34	43	54	60	58	57	53	49	43	40	34	31	
All-Sky Solar Radiation	RadAvg	859	1060	1348	1554	1680	1491	1657	1699	1324	1041	846	763	
	RadStd	56	76	88	159	178	142	220	146	116	108	73	53	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (40 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page